

The Honest Harvest

Blockchain-Based Food Supply Chain Tracker

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Introduction



The Honest Harvest Supply
Chain

The "Trust Gap"



Normal Databases

They are **extremely fast**, but they rely on **blind trust**.

Anyone with admin access can secretly change an expiration date or ingredient list.



Pure Blockchains

They are **totally unhackable**, but saving huge files is way too **slow and expensive**.

Long product histories directly on them is not viable for real businesses.

Our Solution: The Hybrid Model

Honest Harvest gives you the best of both worlds, combining **web speed** with **blockchain security**.



Fast

No waiting for slow networks.
Traditional database speed for
real-time tracking.



Detailed

Can handle complex tracking.
Stores full product histories
without high costs.



Secure

Mathematically impossible to
fake. Every data entry is anchored
by a blockchain hash.

The "Digital Fingerprint"



When you add a product, our system creates a tiny, unique digital fingerprint of all its data (a "Hash").

Digital Fingerprint:

`0xac0974bec39a17e36ba4a6b4d238ff94...`



We lock **only** this fingerprint on the blockchain.

If a hacker alters the database, the fingerprint breaks, exposing the cheat instantly.

Core Architecture



System
Overview

Our Ecosystem

We use [Docker](#) to keep four main tools perfectly in sync:



Frontend

React / Next.js



Backend

Node.js API



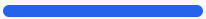
Database

PostgreSQL



Blockchain

Hardhat



How It Works Behind the Scenes



The Family Tree

Our Postgres database does the heavy lifting, tracking exactly how raw ingredients combine to make final products.



Security First

Our Node.js backend acts like a bouncer, ensuring only verified company managers and employees can make changes.



Low Cost

Our backend only talks to the blockchain when absolutely necessary to lock in a new fingerprint, keeping costs near zero.

The Immutable Ledger

The Smart Contract handles the final step.

Instead of saving massive files, the contract simply records the **item ID**, the **owner**, and the **tiny digital fingerprint**.

- ✓ 100% Security
- 🔒 Tamper-proof Records
- 🔄 Zero Performance Lag



Thank You

